

Problem

Rework Prob. 7.17 if $i(0) = 10$ A and $v(t) = 20u(t)$ V.

Step-by-step solution

Step 1 of 6

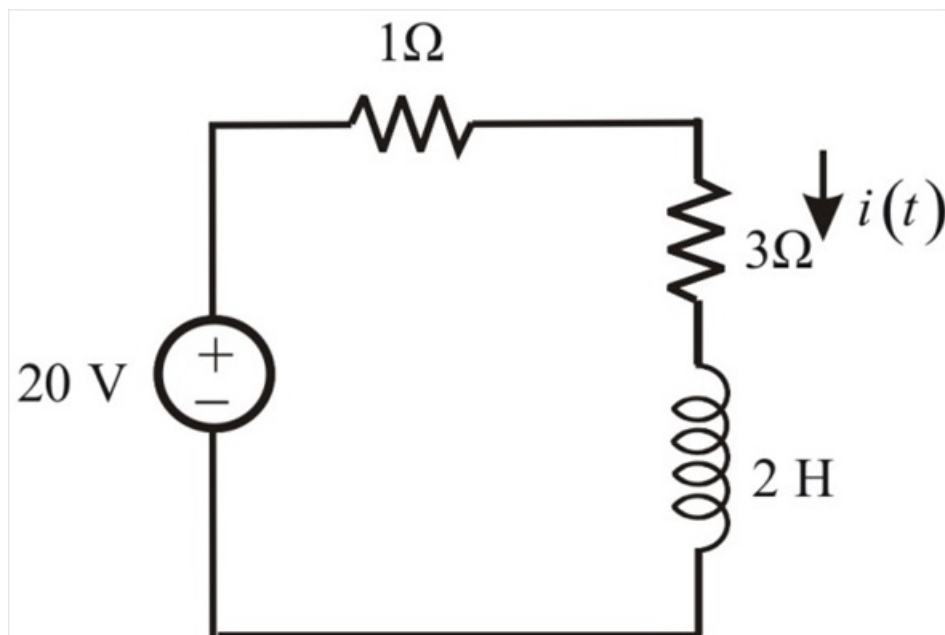
Refer to figure 7.97 in the textbook.

Consider the step input.

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

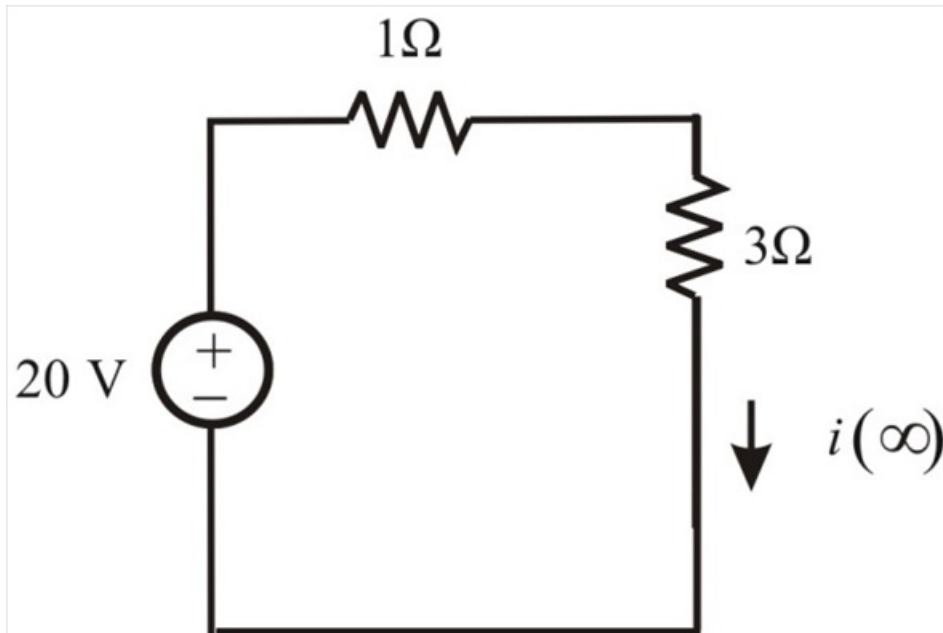
Step 2 of 6

Draw the circuit for $t > 0$ as shown in the following figure:



Step 3 of 6

Consider that the circuit is in that position for a long time and has reached steady state; and inductor acts like a short circuit. Consider the following circuit:



Step 4 of 6

Calculate the steady state current flowing through the inductor.

Apply the Ohm's law.

$$i(\infty) = \frac{20}{1+3}$$

$$i(\infty) = 5 \text{ A}$$

Therefore, the inductor current is,

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$= 5 + [10 - 5]e^{-\frac{t}{\tau}}$$

$$i(t) = 5 \left(1 - e^{-\frac{t}{\tau}} \right) \text{ A}$$

Here, τ is time constant.

Step 5 of 6

Calculate the time constant.

$$\tau = \frac{L}{R_{eq}}$$

Determine the equivalent resistance for $t > 0$.

$$R_{eq} = 3 + 1$$

$$= 4 \Omega$$

Therefore,

$$\tau = \frac{2}{4}$$

$$\tau = \frac{1}{2} \text{ sec}$$

Therefore, the inductor current is,

$$i(t) = 5 \left(1 - e^{-2t} \right) \text{ A}$$

Therefore, the output voltage is,

$$v_o(t) = i(t)R + L \frac{di(t)}{dt}$$

$$\begin{aligned} v_o(t) &= 3 \times 5(1 - e^{-2t}) + 2 \frac{d}{dt}(5(1 - e^{-2t})) \\ &= 15(1 - e^{-2t}) - 2(5)(-2)e^{-2t} \end{aligned}$$

$$v_o(t) = 15 + 5e^{-2t} \text{ V}$$

Therefore, the voltage $v_o(t)$ for $t > 0$ is $\boxed{15 + 5e^{-2t} \text{ V}}$.